

Accuracy Hides Semantic Error: NeSyArena, a Measurement Instrument for Neuro-Symbolic Reasoning

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Abstract

Neuro-symbolic (NeSy) reasoning systems are compared by task accuracy, yet accuracy cannot tell whether a system computes the semantics it claims. We define *semantic fidelity* — the signed, oracle-grounded error between the value a system computes and the value its declared semantics defines — and build NESYARENA, an open diagnostic instrument that measures it under controlled program structure (proof multiplicity, overlap, recursion depth, aggregation surrogates) together with its learning consequences (gradient liveness, calibration against exact Bayes posteriors, held-out transfer). From an algebraic analysis of rule-based NeSy reasoning we derive signed error laws for the common approximations and verify them to machine precision; the laws are *registered predictions*, and the instrument has by now *refuted two of our own registered hypotheses* — the strongest evidence that it measures something derivation alone does not give. Pointing the instrument at a deployed system produced a finding within a day: Scallop’s `addmultprob` under recursion computes proof-sums truncated at the tuple-set fixpoint (neither distribution semantics nor infinitary proof-summing). End-to-end, task accuracy is statistically indistinguishable across five reasoners while calibration and transfer separate them decisively, on synthetic pixels and on real MNIST digits; a CLUTRR-style study shows the truncation horizon reappearing as a systematic-generalization cliff at exactly the predicted length. All generators, oracles, adapters, configurations and seeds are released; every number in this report regenerates with one command.

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1 Reader’s guide, project history, and why we pivoted

This document is written for colleagues who need to understand — and challenge — what this project claims. It is deliberately more complete than a conference paper: every symbol is defined (§2, Appendix A), every law is stated with proof (§4), the methodology that protects the results is described (§5), the limitations are listed without euphemism (§11), and anticipated objections are answered explicitly (§12).

1.1 Where the project comes from

The project began as a theory paper submitted to KR 2026 (paper 305, *A Semantics-First Contract for Neuro-Symbolic Reasoning*). That paper factored rule-based NeSy systems into four degrees of freedom — a truth-value algebra $(V, \oplus, \otimes, 0, 1)$, body composition, derivation aggregation, and a learnable base interpretation f_θ — and proved two correspondence theorems (restated as Theorems 1 and 2 below): bounded unrolling needs only finitary semiring algebra, while unbounded recursion needs completeness and ω -continuity. An “assumption stack” turned this into a checklist for auditing systems.

The submission received three weak rejects. The criticisms, on re-reading, were substantially correct, and the project’s current shape is a direct response to them:

- “*The math is semiring-PLP re-presented*” (Reviewer C). Correct: the theorems are semiring-valued versions of classical results (van Emden–Kowalski unfolding; Kleene fixed point), already implicit in the provenance-semiring and algebraic-model-counting literature. Consequence: the theorems are now *cited background*, never the contribution.
- “*It only explains problems that were already found and fixed*” (Reviewer B). Correct as submitted: a checklist that post-dicts known bugs is not a result. Consequence: the contribution must be an *instrument* whose measurements produce new facts — including facts that contradict its authors’ expectations. This report documents two such refutations (§9).
- “*Almost entirely theoretical*” (Reviewer A). Correct. Consequence: the empirical program of §§6–8.

1.2 The pivot, in one paragraph

The pivot is from *semantics as theory* to *semantics as measurement*. The algebra stops being the claim and becomes the calibration standard: for every system under test we fix the semantics it *claims* to compute, compute that semantics *exactly* with an independent oracle on the identical inputs, and report the signed difference as a function of program structure. The theory’s role is to generate falsifiable, signed, structural predictions (the error laws of §4); the instrument’s role is to confirm, refute, or refine them — and to measure what the theory underdetermines, which turned out to be where the interesting results are.

1.3 Positioning in one sentence

DeepLog, Scallop and Lobster are *substrates* — they execute a chosen semantics, by compilation or by selectable provenance; NESYARENA is *metrology* — it measures whether what a deployed system computes matches what it claims, and what the residual error does to learning. The relationship is that of conformance test suites to compilers; §10 develops this and §6 shows both kinds of substrate measured by the instrument.

2 Formal setting

Everything in this section is standard or comes from the KR submission; it is the language the instrument is built in. Appendix A tabulates all notation.

2.1 Programs

Definition 2.1 (Signature, atoms, Herbrand base). *A signature Σ consists of finitely many constants and predicate symbols. A (ground) atom is $p(c_1, \dots, c_k)$ with p a k -ary predicate and c_i constants. The Herbrand base $\text{HB}(\Sigma)$ is the set of all ground atoms over Σ .*

Definition 2.2 (Ground program; EDB/IDB). *A ground rule is $h \leftarrow b_1 \wedge \dots \wedge b_m$ with $h, b_i \in \text{HB}(\Sigma)$. A ground program $\mathcal{P}$ is a finite set of ground rules. A predicate is *intensional (IDB)* if it occurs in some rule head, else *extensional (EDB)*. We write q for a distinguished query atom.*

We work with ground programs throughout (the implementation grounds non-ground families during generation); the positive (negation-free) fragment is the scope — see §11.

2.2 Algebras

Definition 2.3 (Commutative semiring). *$(V, \oplus, \otimes, 0, 1)$ where $(V, \oplus, 0)$ and $(V, \otimes, 1)$ are commutative monoids, $\otimes$ distributes over $\oplus$, and 0 is absorbing for $\otimes$. We read $\otimes$ as composition along a rule body and $\oplus$ as aggregation over alternative derivations. $\oplus$ is idempotent if $a \oplus a = a$; idempotent $\oplus$ induces an order $a \leq b \Leftrightarrow a \oplus b = b$, making $\oplus$ a join.*

Instances used in this report: **BOOLEAN** $(\{0, 1\}, \vee, \wedge)$ (we implement it as $(\max, \min)$ on $[0, 1]$, i.e. Gödel), **MAXPROD** $([0, 1], \max, \cdot)$, **SUMPROD** $(\mathbb{R}_{\geq 0}, +, \cdot)$, **TROPICAL** $(\mathbb{R}_{\geq 0} \cup \{\infty\}, \min, +)$.

Definition 2.4 (Quantale; ω -continuity). *A commutative quantale is $(V, \leq, \otimes, 1)$ where $(V, \leq)$ is a complete join-semilattice (every subset has a supremum $\bigvee$) and $\otimes$ distributes over arbitrary joins. An endofunction F on a poset is ω -continuous if $F(\bigvee_n x_n) = \bigvee_n F(x_n)$ for every increasing ω -chain (x_n) .*

2.3 Interpretations, the operator view, the proof view

Definition 2.5 (Interpretations; base interpretation). *A V -interpretation is $I : \text{HB}(\Sigma) \rightarrow V$, ordered pointwise when V is. The base interpretation $f_\theta : \text{HB}(\Sigma) \rightarrow V$ supplies values for EDB atoms (in NeSy: classifier outputs used as facts, with parameters θ); by convention $f_\theta(a) = 0$ for IDB atoms a , so derived atoms get their value from inference alone.*

Definition 2.6 (Immediate consequence; bounded iteration). *For an interpretation I , body evaluation is $\text{Eval}_I(b_1 \wedge \dots \wedge b_m) = I(b_1) \otimes \dots \otimes I(b_m)$, and*

$$(T_{\mathcal{P}}I)(a) = \bigoplus_{(a \leftarrow \beta) \in \mathcal{P}} \text{Eval}_I(\beta), \quad F_{\mathcal{P}}^{f_\theta}(I) = f_\theta \oplus T_{\mathcal{P}}(I) \quad (\text{pointwise}),$$

with $\bigoplus \emptyset = 0$. Bounded iteration: $I^{(0)} = f_\theta$, $I^{(n+1)} = F_{\mathcal{P}}^{f_\theta}(I^{(n)})$.

Definition 2.7 (Proof trees; depth; value). *A proof tree π for q is a finite tree with root labeled q ; each internal node labeled a has children $b_1, \dots, b_m$ for some rule $a \leftarrow b_1 \wedge \dots \wedge b_m \in \mathcal{P}$; leaves are EDB atoms (or unexpanded atoms, which carry value 0 and can be ignored). $\text{depth}(\pi)$ is the maximum root-to-leaf distance (0 for a single node). The value is $\text{val}_{f_\theta}(\pi) = \bigotimes_{\ell \in \text{leaves}(\pi)} f_\theta(\ell)$, with leaf multiplicity (a fact used by two subtrees contributes twice under non-idempotent $\otimes$). The support $\text{supp}(\pi)$ is the set of its EDB leaves. Depth-bounded proof aggregation: $\text{Infer}^{(n)}(q) = \bigoplus_{\pi : \text{depth}(\pi) \leq n} \text{val}_{f_\theta}(\pi)$.*

2.4 The two correspondence theorems (cited background)

Theorem 1 (Bounded correspondence; KR submission Thm. 1). *Assume (i) finitely many ground rule instances per head atom, and (ii) $(V, \oplus, \otimes, 0, 1)$ a commutative semiring. Then for every $n \in \mathbb{N}$ and ground q : $I^{(n)}(q) = \text{Infer}^{(n)}(q)$. Only finitary algebra is needed.*

Theorem 2 (Unbounded correspondence; KR submission Thm. 2). *Assume additionally that $(V, \leq, \otimes, 1)$ is a commutative quantale with $\oplus = \vee$ and that $F_{\mathcal{P}}^{f_\theta}$ is ω -continuous (which holds under countably many rule instances per head). Then $I^* := \text{lfp}(F_{\mathcal{P}}^{f_\theta})$ exists by the Kleene fixed-point theorem and $I^*(q) = \bigvee_n \text{Infer}^{(n)}(q)$ for every q .*

Remark 1 (Attribution and status). These are semiring-valued forms of classical results (van Emden–Kowalski 1976; Lloyd 1987; Kleene) in the lineage of provenance semirings (Green et al. 2007), algebraic model counting (Kimmig et al. 2017), semiring Datalog convergence (Khamis et al. 2024) and Bourgaux et al. (2022). We do not claim them; we *verify them numerically inside the instrument*: operator iterates equal proof aggregation, with proofs *enumerated mechanically from the program IR*, to 10^{-12} on a cyclic program under non-idempotent $\oplus$ (§5). The gap between Theorem 1 (cheap) and Theorem 2 (expensive assumptions) is exactly where deployed systems improvise — and where Finding 1 lives.

Definition 2.8 (Distribution semantics; WMC). *Let the EDB atoms $f_1, \dots, f_m$ be independent Bernoulli facts with $\Pr(f_i) = p_i \in [0, 1]$. A world $w \in \{0, 1\}^m$ has probability $\pi(w) = \prod_i p_i^{w_i} (1 - p_i)^{1-w_i}$. For a monotone query q whose proofs have supports $\sigma_1, \dots, \sigma_P \subseteq \{f_1, \dots, f_m\}$, the distribution-semantics value is the weighted model count*

$$P^*(q) = \Pr\left(\bigcup_j E_{\sigma_j}\right) = \sum_{w : w \models q} \pi(w), \quad E_\sigma = \bigwedge_{f \in \sigma} f.$$

Computing P^ correctly requires a disjoint representation (knowledge compilation / AMC), not proof enumeration: proof-sums double count.*

3 Semantic fidelity: the measurement framework

3.1 Claimed semantics and the central definition

Definition 3.1 (Claimed semantics; oracle). *A system under test (SUT) S declares a claimed semantics: the mathematical object it purports to compute (e.g. “distribution semantics”, “max-join over proof scores”, “least fixed point under $(\max, \cdot)$ ”). The oracle value $I^*(q)$ is that object computed exactly, by an independent method, from the identical base interpretation f_θ the SUT consumes.*

Definition 3.2 (D1: semantic error). $\varepsilon_S(q) = \hat{S}(q) - I^*(q)$, where $\hat{S}(q)$ is what the SUT returns. Signed by design: over- and under-counting are different failure modes with different learning consequences. Because oracle and SUT share f_θ , ε isolates the reasoning layer from perception.

Definition 3.3 (D2: fidelity profile). *For structured sweep families $\{G_a\}$, one per diagnostic axis a (§5.2), $\phi_a(S) = \max(0, 1 - \mathbb{E}_{G_a}|\varepsilon_S|)$. The vector $\phi(S)$ is the scorecard (Figure 4); raw signed errors are always reported alongside.*

Definition 3.4 (D3: depth horizon). *On a family with required proof depth L (chains), $h_\delta(S) = \min\{L : |\varepsilon_S| > \delta\}$. For an n -step unroller, Theorem 1 predicts $h_\delta = n + 1$.*

Definition 3.5 (D4: gradient liveness). *With oracle gradient ∇^* and system gradient $\hat{\nabla}$ (each system differentiates its own value function — see Def. 3.8),*

$$\text{live}(S) = \frac{\#\{f : |\hat{\nabla}_f| > \tau \wedge |\nabla_f^*| > \tau\}}{\#\{f : |\nabla_f^*| > \tau\}},$$

the share of facts the oracle would update that the SUT also updates ($\tau = 10^{-12}$ for analytic gradients, 10^{-9} in batched settings).

Definition 3.6 (D5: learning-consequence metrics). *In learning experiments the perception module is trained through each SUT and evaluated against exact ground truth: calibration = mean $|p_{\text{percept}} - p_{\text{Bayes}}|$ where the Bayes posterior is closed-form by construction of the generative model; transfer = mean $|\hat{S}(q_h; \text{learned } f_\theta) - I^*(q_h; \text{true } f_\theta)|$ over held-out queries q_h , i.e. deployment error of the whole trained pipeline; facts-quality = the same with the exact reasoner substituted at deployment, isolating perception corruption from deployment-time reasoning error. The oracle floor is the exact-trained pipeline’s value: irreducible estimation error.*

Definition 3.7 (D6: witness). *A minimal generator configuration g with $|\varepsilon_S(g)| > \delta$, found by ordered grid search followed by greedy shrinking along each parameter (QuickCheck-style). Witnesses are the machine-found analogue of hand-crafted failure micro-examples.*

3.2 The reference systems under test

Definition 3.8 (Reference SUTs and their gradient semantics). *Let q have proof supports $\sigma_1, \dots, \sigma_P$ and scores $s_j = \prod_{f \in \sigma_j} p_f$. The five reference SUTs, each mirroring a deployed aggregation strategy including its differentiation behavior:*

- **exact-wmc**: $P^*(q)$ (Def. 2.8); gradient analytic (Prop. 5). Claimed semantics: distribution semantics; $\varepsilon \equiv 0$ by definition.
- **add-mult(clamped)**: $\min(1, \sum_j s_j)$. Gradient: $\partial/\partial p_f \sum_j s_j$ when the sum is < 1 ; identically zero when ≥ 1 (the clamp’s flat region — “gradient blackout”). Claim: distribution semantics.
- **top-k**: P^* restricted to the k highest-scoring proofs; the selection is frozen (not differentiated through), gradients flow through the restricted WMC (difftopkproofs-style). Claim: distribution semantics.
- **min-max**: $\max_j \min_{f \in \sigma_j} p_f$; subgradient is one-hot at the bottleneck fact of the best proof (ties broken lexicographically — a deliberate, documented determinism fix over the toy, whose tie depended on hash order). Claim: distribution semantics.
- **LSE $_\tau$** : $\tau \log \sum_j e^{s_j/\tau}$, the smooth surrogate for the max-join; its claimed semantics is $\max_j s_j$ (so its oracle is the max, not WMC); softmax-weighted gradients.

Remark 2 (Why reference SUTs and not only deployed systems). The reference SUTs are exactly specified, fast, and differentiable with controlled semantics; the deployed systems are then *measured against them* (§6). The measured result — agreement at machine precision on the acyclic battery — is what authorizes transferring every reference-SUT result to the deployed systems. Where deployed behavior *deviates* from the reference model (recursion; Finding 1), the deviation itself is the result.

4 Error laws

Each law converts one violated algebraic commitment into a *signed, registered prediction*. They are deliberately elementary; their value is falsifiability, not depth. All were verified to the stated precision by the rebuilt instrument.

Proposition 1 (Proof-sums over-count; clamp blacks out). *With events $E_j = E_{\sigma_j}$ (Def. 2.8),*

$$0 \leq \sum_j s_j - P^*(q) \leq \sum_{i < j} \Pr(E_i \wedge E_j),$$

with equality on the left iff the E_j are pairwise disjoint. Moreover $\min(1, \sum_j s_j)$ has identically zero gradient wherever $\sum_j s_j \geq 1$.

Proof. $s_j = \Pr(E_j)$ under independent facts. The union bound gives $P^* = \Pr(\bigcup E_j) \leq \sum_j \Pr(E_j)$, i.e. the left inequality; Bonferroni's second-order lower bound gives $\Pr(\bigcup E_j) \geq \sum_j \Pr(E_j) - \sum_{i < j} \Pr(E_i \wedge E_j)$, i.e. the right. Equality on the left iff no overlap mass, i.e. pairwise disjointness. The clamp claim is immediate: $x \mapsto \min(1, x)$ is constant on $[1, \infty)$. *Status: verified; the Bonferroni upper bound is asserted in the standing test battery; witness +0.160 at $(P, L, c, p) = (2, 1, 0, 0.6)$.* $\square$

Proposition 2 (Top- k under-counts by at most the dropped mass). *Let K be the set of k highest-scoring proofs. Then*

$$0 \leq P^*(q) - \Pr\left(\bigcup_{j \in K} E_j\right) \leq \sum_{j \notin K} s_j,$$

zero iff every dropped proof is covered (almost surely) by the kept union; in particular $\varepsilon = 0$ whenever $P \leq k$.

Proof. Monotonicity of probability gives the left inequality. For the right: $\Pr(\bigcup_{\text{all}}) - \Pr(\bigcup_K) \leq \Pr(\bigcup_{j \notin K} E_j) \leq \sum_{j \notin K} \Pr(E_j)$. *Status: verified; witnesses -0.210 ($k=1$) and -0.103 ($k=3$); the $k \geq P$ identity is additionally measured end-to-end in both learning controls (§8).* $\square$

Proposition 3 (Surrogate bias law). *For scores $s_1, \dots, s_P$, $P \geq 2$: $\max_j s_j < \text{LSE}_\tau(s_1, \dots, s_P) \leq \max_j s_j + \tau \ln P$, with equality on the right iff all scores are equal; for equal scores the bias is exactly $\tau \ln P$.*

Proof. Write $M = \max_j s_j$. Strict lower bound: $\sum_j e^{s_j/\tau} > e^{M/\tau}$ since the other $P-1$ terms are positive. Upper bound: $\sum_j e^{s_j/\tau} \leq P e^{M/\tau}$, so $\text{LSE}_\tau \leq M + \tau \ln P$, with equality iff every $s_j = M$. *Status: verified to 2.22×10^{-16} over a (P, τ) grid.* $\square$

Proposition 4 (Truncation starves gradients). *If the minimal proof depth of q exceeds the unrolling budget n (and f_θ assigns 0 to IDB atoms), then $I^{(n)}(q) = 0$ and $\nabla_\theta I^{(n)}(q) \equiv 0$.*

Proof. By Theorem 1, $I^{(n)}(q) = \text{Infer}^{(n)}(q)$, and the depth- $\leq n$ proof set is empty, so $I^{(n)}(q) = \bigoplus \emptyset = 0$ identically in a neighborhood of θ : perturbing fact values creates no proof of depth $\leq n$. A constant function has zero gradient. *Status: verified — value and finite-difference gradient exactly 0 below the horizon; measured horizons equal $n+1$ for $n \in \{2, 4, 6, 8, 10\}$; end-to-end consequences in E7 and E8.* $\square$

Proposition 5 (Analytic WMC gradient). *For $P^*(q)$ as in Def. 2.8 and any fact f with $p_f \in (0, 1)$:*

$$\frac{\partial P^*(q)}{\partial p_f} = \Pr(q \mid f) - \Pr(q \mid \neg f) = \sum_{w: w \models q} \pi(w) \frac{w_f - p_f}{p_f(1 - p_f)}.$$

Proof. Split the worlds by w_f : $P^* = p_f \Pr(q \mid f) + (1 - p_f) \Pr(q \mid \neg f)$, where the conditional terms do not depend on p_f ; differentiate. The second form follows since $\pi(w)/p_f$ (resp. $\pi(w)/(1 - p_f)$) is the conditional world probability. *Status: matches finite differences to 10^{-6} (step 10^{-7}) and is the oracle for all gradient-conformance comparisons.* $\square$

Remark 3 (What the laws do *not* determine). The laws bound each error in isolation. They do not determine (i) the crossover point where the $|\varepsilon|$ -ranking of over- and under-counting flips (measured: between $c=1$ and $c=2$ at $P=4, L=4, p=0.6$); (ii) the sign of min-max error (measured: structure-dependent, $+0.39$ vs -0.47 in the toy study); (iii) any of the learning-time interactions of §8. That gap is the argument for the instrument.

5 The instrument and its methodology

5.1 Architecture

The rebuilt codebase (`src/nesyarena/`, 107 tests) is organized as:

<code>ir.py</code>	ground-program IR: <code>Atom/Rule/GroundProgram</code> ; bounded proof enumeration
<code>algebra.py</code>	semiring registry (<code>BOOLEAN</code> , <code>MAXPROD</code> , <code>SUMPROD</code> , <code>TROPICAL</code>)
<code>engine.py</code>	T_P iteration, run-to-convergence, proof-side aggregation
<code>oracle.py</code>	brute-force WMC ($m \leq 22$) + analytic gradients; ProbLog oracle; graph oracles
<code>suts.py</code>	the five reference SUTs of Def. 3.8
<code>generators.py</code>	program families G1–G3 (§5.2)
<code>metrics.py, witness.py</code>	D2–D4, D6
<code>learning/</code>	batched torch provenance layer whose <i>plain autograd</i> is system-faithful
<code>adapters/</code>	frozen adapter protocol; Scallop, DeepLog

Programs are the primary object: generators emit programs, proof sets are *derived* by bounded enumeration (dual view: leaf multisets for semiring evaluation; support sets as the DNF for WMC), and external adapters compile the same object. One documented caveat: enumeration identifies proof trees by leaf multiset; two distinct trees with identical multisets would collapse, which matters only under non-idempotent $\oplus$ and cannot occur on the families used (one rule per proof; transitive closure, where derivations are walks with unique edge multisets).

5.2 Generators (exact specifications)

- **G1 (overlap).** Parameters (P, L, c, p) , optionally heterogeneous: query q with P proofs, each of length L , sharing c “trunk” facts; private facts $x_{j,i}$; all probabilities p , or i.i.d. $U[0.35, 0.9]$. Fact count $m = c + P(L - c) \leq 22$. Axes: multiplicity (P), overlap (c), saturation (p high).
- **G2 (depth).** Chains $v_0 \rightarrow \dots \rightarrow v_L$ with edge probability p , transitive closure program, query $\text{path}(v_0, v_L)$ (minimal proof depth L); the cyclic instance $a \leftrightarrow b \rightarrow c$ ($p_{ab} = p_{ba} = 0.9$, $p_{bc} = 0.8$) as the canonical recursion stress (infinitely many proofs).
- **G3 (surrogate).** Score multisets as single-fact proofs: equal scores (s, P) for the bias law; one boosted proof $(s + \Delta)$ for the dilemma curve.
- **E8 generator (CLUTRR-style lineal kinship).** Relations = (gender, generation offset) with offset in $[-2, 2]$ (10 relations; 6 base relations with $|d| \leq 1$); composition $(g_1, d_1) \circ (g_2, d_2) = (g_2, d_1 + d_2)$, deterministic and closed; chains sampled with all prefix offsets in range.

5.3 Oracles and the validation chain

The distribution-semantics oracle enumerates all 2^m worlds ($m \leq 22$) and computes P^* and its analytic gradient (Prop. 5) in one vectorized pass; beyond $m = 22$, ProbLog’s exact inference (knowledge compilation) is the oracle. Idempotent algebras on transitive closure use direct graph algorithms. *The oracle never rests on itself*: the standing test battery asserts brute-force $\equiv$ ProbLog to $< 10^{-10}$ on random instances (measured 6.7×10^{-16}), analytic $\equiv$ finite-difference gradients, and Theorem 1 numerically on the cyclic program under SUMPROD for $n = 0..10$ with proofs enumerated from the IR.

5.4 Methodological safeguards

1. **Registered predictions.** Every error law and every hypothesis carries its predicted sign and growth direction *written before the run* (in configs and docstrings); outcomes are recorded either way. The ledger (§9) contains two refutations of our own predictions — the discipline is load-bearing, not decorative.
2. **Parity gates.** The first implementation (the “toy”, now `legacy/`) was verified, then frozen as an *executable specification*: golden fixtures pin its oracle values, gradients, engine trajectories and witnesses; the

rebuilt core must reproduce them (tolerances 10^{-10} – 10^{-14}). Learning experiments were re-implemented on torch autograd and reproduce the toy’s numpy results *to every printed decimal including standard deviations* — two independent implementations following identical trajectories.

3. **Shared inputs.** Oracle and SUT always consume the identical $(\mathcal{P}, f_\theta)$; in learning runs all SUTs share data, seeds, and initializations within a seed.
4. **No silent coverage loss.** Sweep cells skipped (e.g. over the WMC fact limit) are logged in the output; configs are committed YAML whose sha256 is embedded in every result file.
5. **Adapters never normalize.** A deployed system’s deviation from its claim must pass through the adapter verbatim; deviations are findings, pinned by canary tests that flip if upstream changes (versions pinned).
6. **One-command reproduction.** `make all` regenerates every number and figure in this report (Appendix B).

6 Conformance of deployed systems

Two deployed backends were measured. Versions: scallopy 0.2.4 (official release wheel); pydeeplog 3.0.3 (PyPI; repo ML-KULeuven/deeplog, last push 2026-06-03); ProbLog 2.2.10.

6.1 Scallop: value and gradient conformance, then a recursion finding

On 50 G1 instances (homogeneous and heterogeneous), Scallop’s provenances match the reference SUTs at machine precision while deviating from the exact oracle by the magnitudes the laws predict:

provenance	$\max \text{scallop} - \text{reference} $	$\max \text{scallop} - \text{oracle} $
addmultprob	2.2×10^{-16}	0.609
minmaxprob	0.0	0.263
topkproofs($k=1$)	5.6×10^{-17}	0.378
topkproofs($k=3$)	2.2×10^{-16}	0.061

The same conformance holds through full program compilation (the IR adapter), on chains run by Scallop’s native recursion (agreement 0.0 with the convergent fixpoint), and at the *gradient* level: central finite differences through Scallop match the references’ system-faithful gradients to 7.9×10^{-12} on tie-free instances — confirming frozen top- k selection and the one-hot min-max subgradient as deployed differentiation behavior. Deployed Scallop exposes no bounded-iteration mode through scallopy 0.2.4 (run-to-fixpoint only), so the depth axis applies to it only via program rewriting.

Finding 1 (Scallop’s `addmultprob` under recursion is not proof-summing). On the cyclic instance, the naive model of add-mult — sum all (bounded-depth) proof scores, clamp at 1 — predicts 1.0; deployed Scallop returns 0.72. A designed probe isolates the mechanism (diamond $a \rightarrow b \rightarrow c$, $a \rightarrow d \rightarrow c$ with 0.6 edges, $\pm$ a back-edge $c \rightarrow a$): on the DAG, Scallop returns exactly the simple-path proof-sum (0.720 vs oracle 0.5904 — the predicted rung-P over-count); adding the cycle changes *nothing*. Derivations that re-derive an already-present tuple never accumulate into its tag: iteration stops at the *tuple-set* fixpoint, freezing tags at their acyclic-derivation values. The deployed semantic object under recursion is therefore *iteration-policy-dependent* — neither distribution semantics nor infinitary proof-summing. Two consequences: (i) on cyclic reachability this accidentally *protects* addmultprob from divergence/saturation (a coincidence of subsumed loop supports, not a guarantee); (ii) the reference AddMult remains a faithful model of deployed behavior on acyclic structure — where the overlap axis lives — but recursive sweeps require a truncated-at-tuple-fixpoint reference variant (not yet implemented; §11).

6.2 DeepLog: conformance pass, including gradients

DeepLog compiles formula specifications to algebraic circuits and claims exactness. Registered prediction: fidelity 1.0 at the circuit’s working precision. Measured: the weighted-model-count construction (assignment-aware $p(F)$ weights $\times$ boolean-to-probability transform of the proof DNF) matches the WMC oracle to $< 10^{-6}$ on the G1 battery (float32 circuits); recursion handled by the arena’s bounded unrolling is exact. Gradients: symbolic (non-numeric) probability labels survive DeepLog’s label resolver and surface as module inputs, so torch autograd through the compiled circuit yields DeepLog’s *deployed* gradients — matching the analytic oracle gradient to 1.3×10^{-7} . A conformance *pass* is as much a result as a deviation: it is what makes the leaderboard contrast (exact substrate vs approximate provenances) a measurement rather than an assertion.

Remark 4 (Why the finding matters beyond bug-hunting). F-1 is an *implicit semantics choice* (iteration policy under recursion), invisible to accuracy-style evaluation and found within hours of pointing the instrument at the system. This is the thesis demonstrating itself: claimed semantics must be measured, not assumed.

7 Inference-level results (E1–E4)

E1 — structural crossover. On the G1 grid (1030 cells; skipped cells logged), add-mult’s signed error grows with overlap ($+0.092 \rightarrow +0.308$ as $c : 0 \rightarrow 3$ at $P=4, L=4, p=0.6$) while top-1’s shrinks ($-0.296 \rightarrow -0.081$): opposite monotone structural sensitivities, with the $|\varepsilon|$ -ranking flipping between $c=1$ and $c=2$ (Fig. 1). *No method dominates; structure decides.* Grid-level fidelity ϕ_{overlap} : exact 1.0; top-3 0.970; add-mult 0.863; top-1 0.857; min-max 0.823.

E2 — depth horizons and recursion stress. Measured horizons are exactly $n+1$ for $n \in \{2, 4, 6, 8, 10\}$ (none within range for $n \geq 12$, logged as such); below the horizon the value is flat zero (finite-difference liveness 0 on every $(n, L=n+1)$ cell); at the horizon, values match theory to 5×10^{-7} ($0.430467 = 0.9^8$, $0.478297 = 0.9^7$). On the cyclic instance, run-to-convergence SUMPROD reaches $3.7895 = 0.72/0.19 > 1$ — the H7 “probability” leaving $[0, 1]$ — while MAXPROD correctly converges to 0.72 (Fig. 2).

E3 — the surrogate dilemma. The bias law holds to 2.22×10^{-16} across (P, τ) ; plotting bias against the gradient share reaching non-maximal proofs shows the unreachable corner: no temperature achieves both low bias and live gradients (Fig. 3).

E4 — witnesses. The synthesizer, given only “find the smallest failing configuration”, returns: add-mult $(P, L, c, p) = (2, 1, 0, 0.6)$, $\varepsilon = +0.160$ (re-deriving the canonical double-counting micro-example from the literature); top-1 $(2, 1, 0, 0.3)$, -0.210 ; top-3 $(4, 1, 0, 0.3)$, -0.103 ; min-max $(1, 2, 0, 0.3)$, $+0.210$ — notably min-max fails on a *single* proof of two facts: it does not need alternative derivations to be wrong. Exact-WMC has no witness on the grid.

Gradient liveness ladder (radar axis). exact 1.000; top-3 0.832; add-mult 0.611 (clamp blackout); top-1 0.498; min-max $0.166 \approx 1/m$ (the one-hot subgradient keeps a single fact alive).

8 Learning-level results (E5–E8)

The learning layer implements each SUT as a batched torch operation whose *plain autograd* reproduces the system-faithful gradients of Def. 3.8 (parity-gated to 10^{-10}): clamp via `torch.clamp`, frozen selection via detached `argsort`, one-hot subgradients via `amin/amax`. Training code is therefore ordinary `loss.backward()`, and each reasoner’s distortions reach the perception module exactly as deployed systems deliver them.

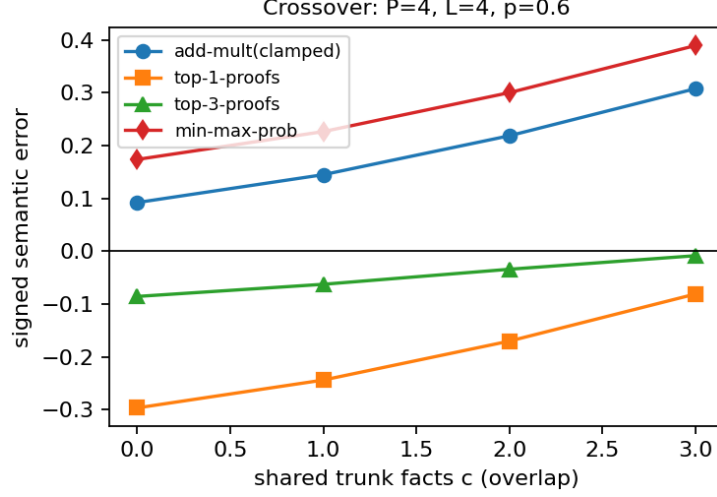


Figure 1: E1: opposite structural sensitivities of over- and under-counting approximations ($P=4$, $L=4$, $p=0.6$); the ranking flips between $c=1$ and $c=2$.

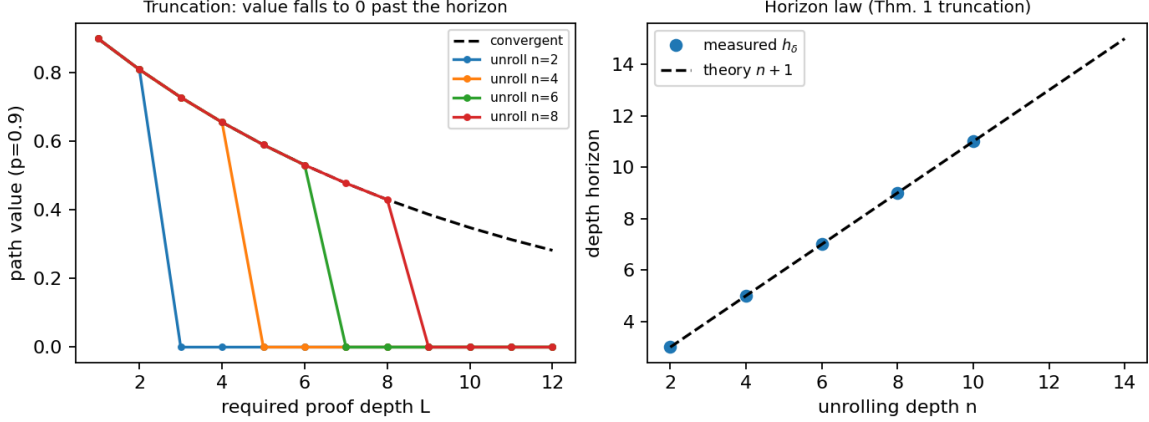


Figure 2: E2: truncation values fall to flat zero past the horizon (left); measured horizons sit exactly on the $n+1$ line (right).

8.1 E6, fact-table setting (population-loss limit; 5 seeds)

A free fact table $\theta \in \mathbb{R}^m$ (probabilities $\sigma(\theta)$) is trained through each reasoner. *Treatment*: G1(4, 3, 2) with heterogeneous ground-truth probabilities; supervision is a 7-query battery (the 6 proof-pairs and the full union) with exact-WMC targets; evaluation is held-out queries (4 singletons + one triple) against exact ground truth, aggregated by the *same* reasoner at deployment. *Control*: the categorical-sum structure (MNIST-sum₂ shape, $|\text{dom}| = 3$): mutually exclusive proofs, where Prop. 1 predicts add-mult is *exactly* distribution semantics; exact and add-mult share one code path so the identity is structural.

SUT	control (disjoint)	treatment (overlap)	facts-quality	fact MAE
exact	0.010 ± 0.020	0.000 ± 0.000	0.000	0.029
add-mult	0.010 ± 0.020	0.101 ± 0.029	0.123	0.128
top-1	0.148 ± 0.076	0.109 ± 0.052	0.113	0.191
top-3	0.010 ± 0.020	0.004 ± 0.003	0.004	0.041
min-max	0.071 ± 0.059	0.095 ± 0.045	0.212	0.272

Reading: training through a misreasoner corrupts the *learned facts themselves* (fact MAE 0.13–0.27 vs

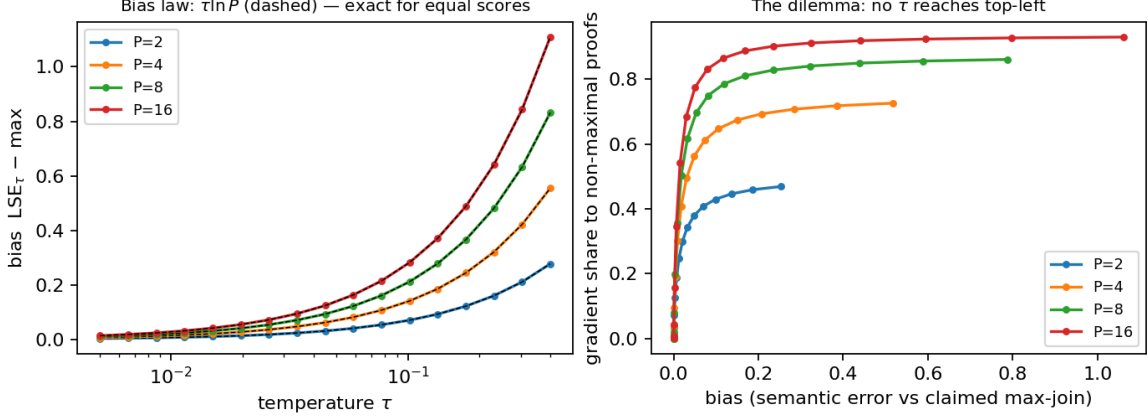


Figure 3: E3: the $\tau \ln P$ bias law (left, dashed = closed form) and the temperature dilemma (right): no τ reaches the low-bias/live-gradients corner.

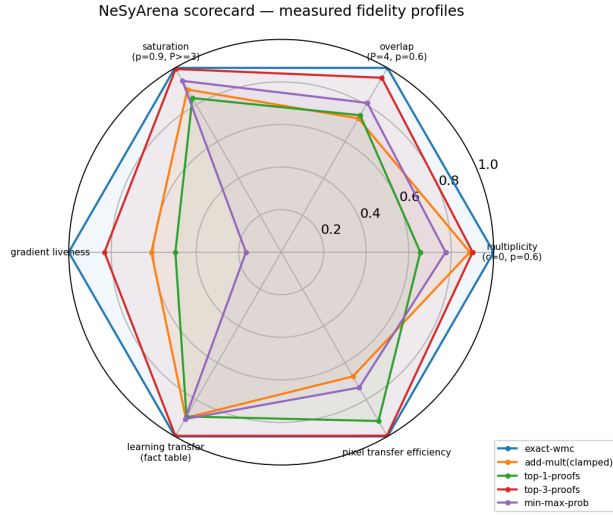


Figure 4: Measured fidelity profiles ϕ over six axes (multiplicity, overlap, saturation, gradient liveness, fact-table transfer, pixel transfer efficiency).

0.03), not just deployment values; on the disjoint control, double-counting is invisible (exact = add-mult) — precisely the cell of the structure grid where the field’s default benchmark lives — while truncation (top-1) is what such tasks *can* see. This is the benchmark autopsy (H9) as a learning experiment.

8.2 E6, pixel perception (the H8 headline; 3 seeds)

Perception is an MLP on synthetic 8×8 images whose generative model is known, so the Bayes posterior is closed-form and calibration is measured against exact ground truth (Def. 3.6). Trained end-to-end through each reasoner on sampled binary labels (Fig. 5):

SUT	accuracy	calibration vs Bayes	transfer	facts-quality
exact	0.740 ± 0.004	0.101 ± 0.003	0.062 ± 0.002	0.062 ± 0.002
add-mult	0.734 ± 0.006	0.145 ± 0.002	0.093 ± 0.002	0.101 ± 0.002
top-1	0.744 ± 0.005	0.105 ± 0.004	0.068 ± 0.003	0.069 ± 0.003
top-3	0.740 ± 0.004	0.101 ± 0.003	0.063 ± 0.003	0.063 ± 0.003
min-max	0.742 ± 0.007	0.128 ± 0.002	0.085 ± 0.004	0.084 ± 0.001

Accuracy is statistically indistinguishable across all five reasoners (0.734–0.744) while calibration and transfer separate them decisively: add-mult sits 44% above the calibration floor and 50% above the transfer floor. On the sum-structured control, $\text{exact} \equiv \text{add-mult} \equiv \text{top-3}$ to every decimal (top-3 measured through the top- k code path), while top-1 is harmed — the autopsy again.

Setting dependence, resolved by a registered ablation (toy study). Top-1, harmful in the fact-table setting, matches the oracle floor under pixel parameterization. We registered the hypothesis (H-A) that supervision richness explains this; the ablation *refuted* it: top-1 ties the floor under both single-query and battery supervision; the operative variable is *parameterization* — a shared perception network absorbs truncation bias that a free fact table amplifies. Add-mult is harmed under both regimes, and decomposes: battery supervision repairs its perception calibration ($0.175 \rightarrow 0.095$) but *not* deployment transfer (0.117), because held-out queries are still aggregated by the faulty reasoner at inference time. (Provenance note: H-A’s numbers are from the verified toy study and are not yet re-run on the rebuilt harness; see §11.)

8.3 E5, MNIST scale (real digits; 3 seeds)

MNIST-path (treatment): a 3×3 grid of MNIST digits; a cell is open iff its digit is even; the query is monotone corner-to-corner reachability — 6 paths of 5 cells, genuinely overlapping. *MNIST-sum* (control): digit pairs from $\{0, 1, 2\}$, train on the sum query (mutually exclusive proofs), transfer to the equality query. Real images have no closed-form Bayes posterior, so this experiment measures the accuracy-vs-transfer divergence (perception quality against true cell states), not exact calibration.

SUT	MNIST-path treatment		
	accuracy	perception MAE	transfer
exact	0.952 ± 0.003	0.097 ± 0.008	0.031 ± 0.001
add-mult	0.934 ± 0.006	0.160 ± 0.016	0.040 ± 0.002
top-1	0.951 ± 0.002	0.095 ± 0.006	0.031 ± 0.001
top-3	0.952 ± 0.004	0.097 ± 0.009	0.031 ± 0.001
min-max	0.940 ± 0.006	0.140 ± 0.006	0.038 ± 0.002

The synthetic findings replicate on real digits: add-mult corrupts perception (+65% MAE over the floor) and transfer (+29%) while accuracy moves by at most ~ 2 points. On the control, $\text{exact} \equiv \text{add-mult} \equiv \text{top-3}$ again — and a *surprise*: top-1 *beats* exact on transfer (0.019 vs 0.221).

8.4 E5b, the knife-edge ablation (registered; one prediction refuted)

We hypothesized (H-N) that top-1’s control win comes from near-deterministic digit posteriors: sharpening acts as a beneficial inductive bias when single-query supervision under-determines perception. Design: inject latent label noise η (cell class = digit label w.p. $1 - \eta$, else uniform), which makes the true posterior closed-form on real digits ($p_{\text{true}}(c) = (1 - \eta)\mathbf{1}[c = \text{label}] + \eta/3$) and restores exact transfer ground truth. Registered: P1 — the advantage decays monotonically in η ; P2 — ranking reverses by $\eta = 0.5$; P3 — $\eta = 0$ anchors at 0.221/0.019.

Measured (Fig. 8): P3 confirmed; P2 confirmed; **P1 refuted** — the advantage *collapses at the first increment* (+0.202 at $\eta=0$ to -0.075 at $\eta=0.15$, then -0.036 , -0.04). Top-1’s superiority on deterministic-perception benchmarks is a knife-edge artifact of exact determinism, not a graded regime. Approximation rankings are regime-dependent in a way none of the laws predict — the second refutation of our own registered prediction, and the strongest version of the “measurement cannot be replaced by derivation” argument.

8.5 E7, depth learning (3 seeds)

On a depth-6 chain task, perception trained through convergent reasoning reaches test AUC 0.722 ± 0.014 ; trained through depth-4 truncated reasoning — whose value below the horizon is flat zero with no gradient path, by Prop. 4, not by simulation — it stays at chance (0.490 ± 0.026) for the entire run (Fig. 6). Gradient starvation, measured end-to-end.

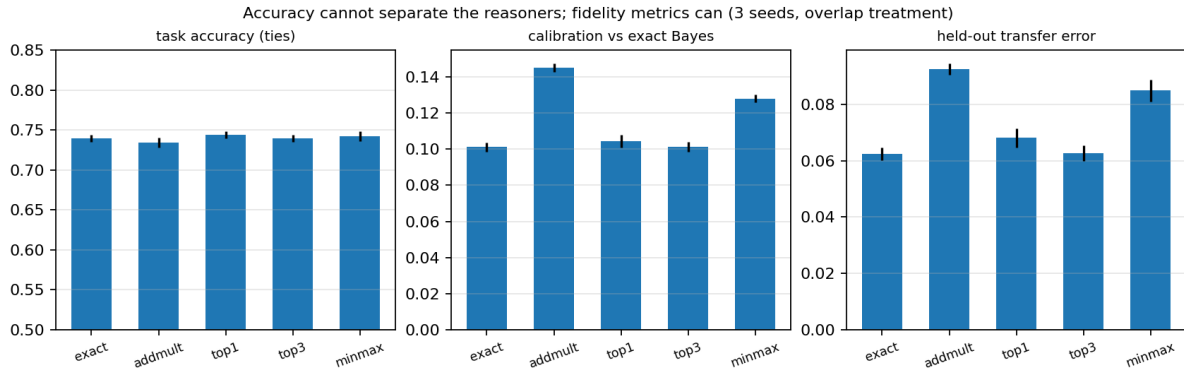


Figure 5: E6 pixels: accuracy cannot separate the five reasoners; calibration against exact Bayes posteriors and held-out transfer can.

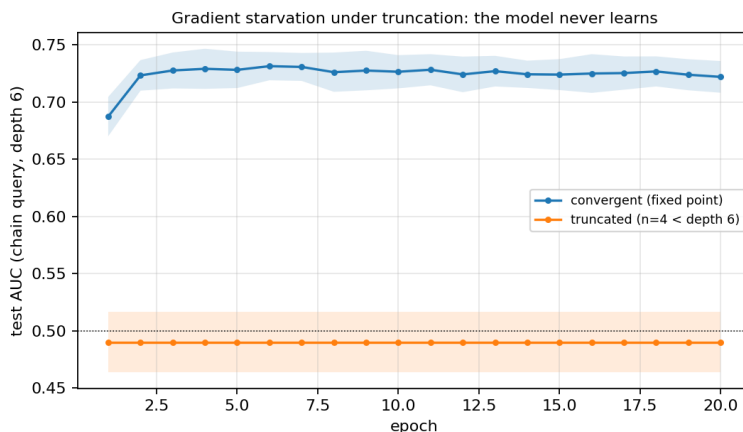


Figure 6: E7: trained through truncated reasoning on a task beyond its horizon, the model never leaves chance.

8.6 E8, CLUTRR-style systematic generalization (3 seeds)

The composition table of the lineal-kinship fragment (§5.2) is itself the learnable object (input embedding U : $6 \rightarrow \Delta^{10}$; table T : $10 \times 6 \rightarrow \Delta^{10}$; sum-product composition along the chain), trained end-to-end on chains of length $k \in \{2, 3\}$ only, tested to $k = 10$ under deployment unroll budgets. Registered: budget- n reasoners are exact for $k \leq n+1$ and collapse beyond; the convergent reasoner generalizes. *Measured* (Fig. 9): cliffs at exactly $k = 4, 6, 10$ for $n = 2, 4, 8$ — with *identical* learned tables across modes — and convergent accuracy 0.73 at $k=10$ (chance 0.10), decaying gracefully as the imperfect learned table (0.885 ± 0.054 of compositions correct) compounds. The truncation horizon of E2/E7 reappears as a systematic-generalization cliff on a community-recognizable task shape.

9 The hypothesis ledger

The registered-prediction discipline, with outcomes. “Toy” marks results verified on the frozen first implementation and not yet re-run on the rebuilt core.

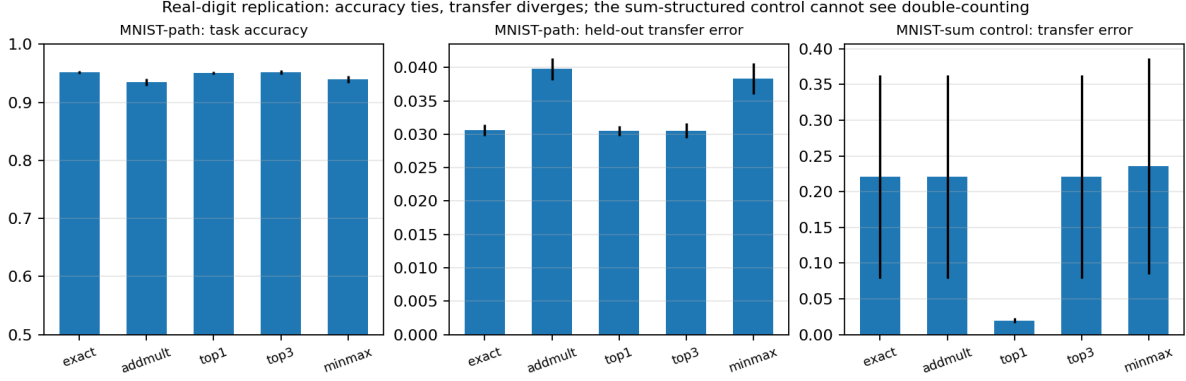


Figure 7: E5 at MNIST scale: transfer diverges while accuracy barely moves; the sum-structured control cannot see double-counting.

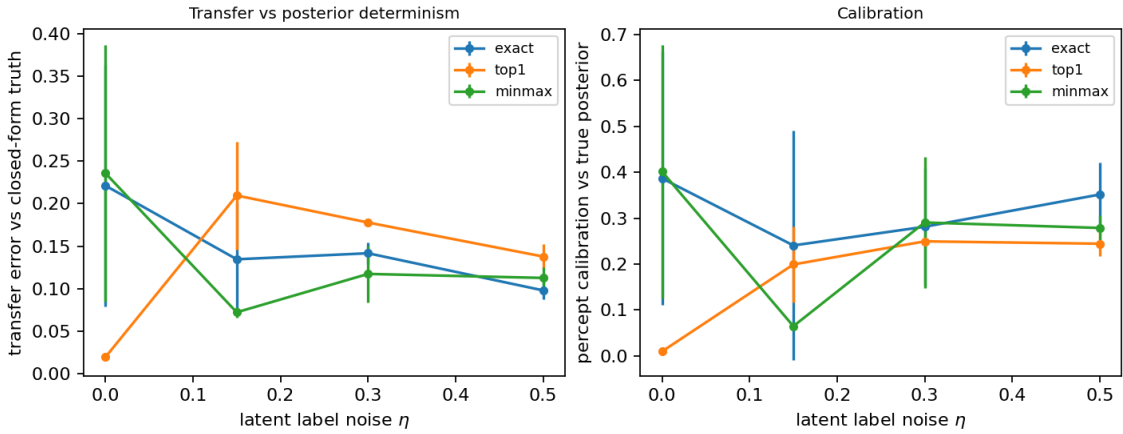


Figure 8: E5b: the knife edge. Top-1's advantage at $\eta = 0$ collapses at the first noise increment (registered prediction P1 refuted).

#	prediction (sign / law)	source	outcome
H1	proof-sum over-counts; $\uparrow$ overlap; Bonferroni bound	Prop. 1	confirmed (E1, battery)
H2	top- k under-counts by dropped mass; 0 iff $P \leq k$	Prop. 2	confirmed (E1, controls)
H3	crossover: ranking flips with overlap	underdetermined	confirmed ($c \in (1, 2)$)
H4	min-max error sign structure-dependent	underdetermined	confirmed (toy)
H5	truncation: 0 value, 0 gradient below horizon $n+1$	Prop. 4	confirmed (E2, E7, E8)
H6	LSE bias = $\tau \ln P$; bias/liveness dilemma	Prop. 3	confirmed (E3)
H7	sum-product on cycles leaves $[0, 1]$ ($\rightarrow 3.79$)	Thm. 2 gap	confirmed (E2)
H8	inference error propagates to calibration/transfer	open	confirmed (E6 both, E5)
H9	MNIST-sum-shaped tasks blind to double-counting	Prop. 1	confirmed (all controls)
H10	non-stratified negation: outside claimed semantics	scope	toy demo only
H-A	top-1's setting dependence due to supervision richness	registered	refuted (toy): parameterization
H-N/P1	top-1 advantage decays monotonically with noise	registered	refuted (E5b): knife-edge collapse
H-N/P2	ranking reverses by $\eta = 0.5$	registered	confirmed (E5b)
F-1	(discovered) addmultprob: tuple-fixpoint truncation	—	measured + probed
F-2	(discovered) diffaddmultprob: straight-through clamp	—	measured + confirmed

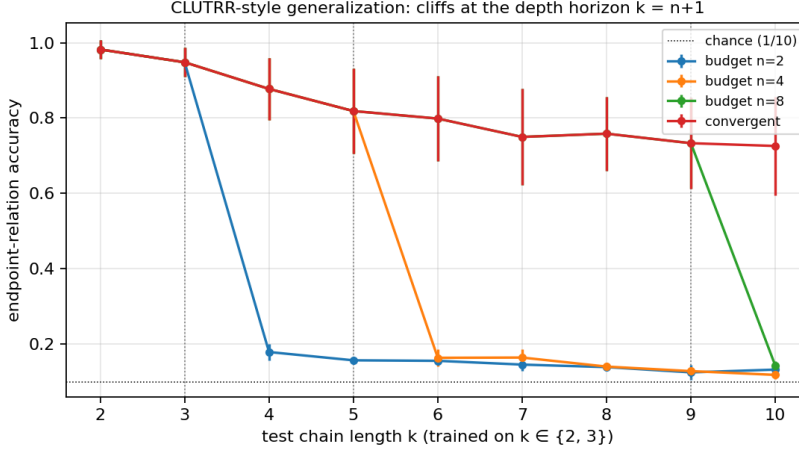


Figure 9: E8: train on $k \leq 3$, test long — generalization cliffs at exactly the depth horizon, with identical learned parameters across budgets.

10 Relation to prior work

Semiring provenance and AMC. Green et al. (2007) introduced provenance semirings; Kimmig, Van den Broeck & De Raedt (2017) separate circuit structure from semiring evaluation and explain when sum-product is correct; Bourgaux et al. (2022) compare semantics for semiring provenance under Datalog recursion; Khamis et al. (2024) analyze convergence of Datalog over (pre-)semirings. Our algebra is inherited from this lineage (Remark 1); none of these works measure deployed systems against claimed semantics.

Substrates. Scallop (Li, Huang & Naik, PLDI 2023) makes the provenance a runtime choice; Lobster (2026) accelerates the same on GPUs; DeepLog (Derkinderen, Manhaeve et al., 2025; software framework 2026) compiles NeSy specifications to algebraic circuits as a universal backend, with DeepProbLog (Manhaeve et al., 2018) as a system layer. Substrates *execute* semantics; they make the question “what does this flag actually compute, and what does it do to learning?” more urgent, not less. We measure all three (§6); the analogy is conformance test suites vs compilers — different artifacts, both infrastructure.

Analyses of NeSy operators and losses. Van Krieken et al. (AIJ 2022) analyze differentiable fuzzy operators; van Krieken et al. (ICML 2024) and Faronius & Dos Martires (2025) debate the independence assumption at the *loss* level, theory-first. Our instrument is model-level and oracle-grounded; the E6/E5 transfer experiments supply controlled-structure evidence that the loss-level debate lacks.

Benchmarks and assurance. rsbench (NeurIPS D&B 2024) targets perception-side reasoning shortcuts — orthogonal to our reasoning-side axis. The PNNL assurance study of Scallop (2025) swept k but measured calibration against accuracy rather than against the exact marginal; our autopsy explains its null result: on sum-structured tasks the dominant failure mode is invisible in principle. CLUTRR (Sinha et al., 2019) and GraphLog instantiate the depth axis in natural-language form; E8 is our controlled stand-in, real-text CLUTRR remains future external validity.

11 Limitations, gaps, and threats to validity

1. **Scope.** Rule/proof-based NeSy with monotone cores. Logic-as-loss systems (semantic loss), non-rule NeSy, and non-monotone semantics (ASP, well-founded) are out of scope; stratified negation exists only as a toy demo (H10) and has *not* been ported to the rebuilt engine.

2. **Recursion coverage of the reference SUTs.** Finding 1 shows deployed add-mult under recursion needs a truncated-at-tuple-fixpoint reference variant; it is specified but not yet implemented, so recursive overlap sweeps against Scallop would currently use a wrong reference model.
3. **Calibration needs synthetic perception.** Exact calibration requires closed-form posteriors; on real images we measure perception MAE and transfer only (E5), and the E5b noise injection is our device for restoring exactness on real data. This is a measurement-design constraint, not an accident.
4. **Statistical depth.** Learning runs use 3 seeds (5 for the fact table); the headline separations are many standard deviations wide, but the protocol’s 5-seed standard is not yet met everywhere. MNIST-sum control variances are large (± 0.14 transfer) because two of three seeds sit in a degenerate optimum — worth more seeds before quoting.
5. **Toy-only results.** The H-A supervision-richness ablation and the H4 sign-dependence numbers are verified on the frozen toy but not yet re-run on the rebuilt core (everything else in this report is rebuilt-core output).
6. **Adapters not yet exercised.** Scallop’s torch-tagged `diff*` provenances (beyond finite differences); DeepLog’s fuzzy/Gödel structures; a standalone (pre-DeepLog) DeepProbLog audit; Lobster.
7. **E8 is CLUTRR-style, not CLUTRR.** Lineal fragment, own generator, no text layer — stated wherever it appears; real-text CLUTRR is future work.
8. **Internal vs external validity.** The toy $\leftrightarrow$ rebuild decimal-exact replication is *internal* (shared seeds and math). External anchors: ProbLog agreement (10^{-16}), Scallop agreement (10^{-16}), DeepLog agreement (10^{-7}), each on independent codebases. Our adapters could still misuse an upstream API; for the Scallop finding we therefore located the mechanism in its evaluation policy and built a minimal witness, rather than resting on black-box disagreement.
9. **The brute-force oracle caps at 22 facts.** Larger instances use ProbLog; learning structures cap at 16 facts (2^m world enumeration). Scaling the oracle (compilation-based) is open.
10. **Generality of generators.** G1–G3 are designed families, chosen to isolate axes; random-program sweeps (random DAGs from the protocol) are not yet productionized.

12 Anticipated questions

Q 1 (“Isn’t this just semiring PLP / the KR paper again?”). The algebra is inherited and explicitly attributed (Remark 1); nothing here re-proves it. What did not exist before: the fidelity definitions (D1–D6), the error laws as *registered* signed predictions, the structure-controlled generators with exact oracles and analytic gradients, the conformance measurements of deployed systems (a finding), and the learning-consequence results. None of these are derivable from the theorems — two of our own registered predictions were refuted by measurement.

Q 2 (“Is this DeepLog? Why is your instrument not redundant given their framework?”). DeepLog executes a chosen semantics by compilation; it does not measure whether a deployed system matches its claim, and it cannot be the neutral referee for its own stack. We measured DeepLog itself: its core passes conformance (including gradients). A substrate and a test suite are different artifacts; the existence of good compilers does not make conformance suites redundant.

Q 3 (“The error laws are trivial.”). Individually, yes — deliberately (we say so in §4). The non-trivial content is (i) what they underdetermine (crossover location, min-max sign, every learning-time interaction), (ii) their conversion into an instrument with exact oracles, and (iii) the measured consequences: accuracy-invisible miscalibration, gradient starvation, benchmark blindness, knife-edge regime dependence.

Q 4 (“Could your finding be a bug in your adapter rather than in the system?”). We (a) reproduced at machine precision the cells where Scallop agrees with its claim (so the adapter is demonstrably faithful elsewhere), (b) designed a minimal witness (the diamond probe isolating tuple-set truncation), and (c) located

the mechanism in the upstream evaluation policy (semi-naive iteration to the tuple-set fixpoint). The finding is a claim about a pinned version, guarded by canary tests.

Q 5 (“Why are the synthetic perception experiments not a weakness?”). Because calibration against *exact* posteriors is the quantity of interest and only a known generative model provides it. The MNIST experiments (E5) show the findings survive real images; the noise ablation (E5b) shows how to recover exact ground truth on real data by owning the stochasticity. The instrument’s claim is about reasoning-layer fidelity, which is exactly what this design isolates.

Q 6 (“Accuracy ties: maybe your tasks are just too easy/hard?”). The ties occur at accuracies far from both chance and ceiling (0.73–0.75 synthetic, 0.93–0.95 MNIST), and the same trained pipelines are separated decisively by calibration and transfer. The mechanism is structural — monotone distortions preserve rankings — not a ceiling artifact.

Q 7 (“Why should anyone adopt this rather than just being careful?”). F-1 occurred in a careful, well-engineered system by a strong group; H-A and H-N/P1 were wrong expectations held by people who had just built the instrument. “Being careful” does not produce signed, oracle-grounded, regression-pinned measurements; a 30-line adapter does (`docs/ADAPTERS.md`).

Q 8 (“What would falsify the headline claim?”). A structured family on which a non-exact aggregation matches exact calibration/transfer while its inference-time $|\varepsilon|$ is large (breaking H8’s propagation), or a deployed system whose measured semantics contradicts our reference model on the acyclic battery (breaking the conformance transfer). Both are exactly what the instrument is built to detect — and publishing them would be a result, as E5b shows.

13 Roadmap

(i) Close the gaps that block claims: truncated-fixpoint reference variant for recursive add-mult; port H-A and negation to the rebuilt core; 5 seeds everywhere; more seeds on the MNIST control. (ii) Externalize: arXiv the theory companion once KR decisions allow; venue per the standing plan (AAAI-27 cycle next). (iii) Extend: Scallop `diff*` gradients via torch tags; DeepLog fuzzy structures; Lobster; real-text CLUTRR; random-DAG sweeps; compilation-based oracle beyond 22 facts.

A Notation

$\Sigma, \text{HB}(\Sigma)$	signature; Herbrand base (Def. 2.1)
$\mathcal{P}; q$	ground program; query atom (Def. 2.2)
$(V, \oplus, \otimes, 0, 1)$	commutative semiring; $\otimes$ body composition, $\oplus$ aggregation (Def. 2.3)
f_θ	base interpretation: EDB values from perception; 0 on IDB (Def. 2.5)
$T_{\mathcal{P}}, F_{\mathcal{P}}^{f_\theta}, I^{(n)}$	consequence operator; fact-augmented operator; n -step iterate (Def. 2.6)
π , depth, val, supp	proof tree, its depth, value, support (Def. 2.7)
$\text{Infer}^{(n)}(q)$	depth-bounded proof aggregation (Def. 2.7)
I^* , lfp	least-fixed-point semantics (Thm. 2)
$P^*(q), \pi(w)$	distribution-semantics value (WMC); world probability (Def. 2.8)
σ_j, s_j, E_j	proof support; score $\prod_{f \in \sigma_j} p_f$; event $\bigwedge_{f \in \sigma_j} f$
P, L, c, p	G1 parameters: #proofs, length, shared trunk, probability
$\hat{S}(q), I^*(q), \varepsilon_S(q)$	SUT value; oracle value under the claim; semantic error (Def. 3.2)
$\phi_a(S), h_\delta$, live	fidelity profile, depth horizon, gradient liveness (Defs. 3.3–3.5)
LSE_τ	$\tau \log \sum_j e^{s_j/\tau}$ (Def. 3.8)
η	latent label-noise rate (E5b)
$k; n$	chain length (E8) / top- k size; unrolling budget

B Reproduction

`python3 -m venv .venv && .venv/bin/pip install -e "[dev,oracles,reporting]", then make all` (tests $\rightarrow$ E1–E4 $\rightarrow$ E6/E7 $\rightarrow$ E5/E5b $\rightarrow$ E8 $\rightarrow$ scorecard + report). Every experiment is a commit-

ted YAML config whose sha256 is embedded in its JSON output; figures regenerate from the JSONs; `out/RESULTS.md` is auto-generated. Version pins for the deployed systems are recorded in the conformance logs (`out/conformance_scallop.md`, `out/conformance_deeplog.md`, `out/conformance_problog_kbest.md`, `out/conformance_out/conformance_ltn.md`). The frozen first implementation lives under `legacy/` with the golden parity fixtures under `tests/fixtures/`.

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